

Prototype Implementation of a Quadrature Down Converter

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Abstract—This paper presents the design, analysis, simulation, and hardware prototype of a Quadrature Down Converter (QDC) operating at a local oscillator frequency of 170 kHz and producing a 5 kHz intermediate frequency (IF) output. The architecture consists of four cascaded stages: (i) a Wien-bridge quadrature oscillator generating 90°-separated sine and cosine signals at 170 kHz, (ii) two NMOS switching mixers multiplying the 175 kHz RF input against each oscillator phase, (iii) passive RC low-pass filters with a −3 dB cutoff near 4.82 kHz for image rejection, and (iv) a common-source NMOS amplification stage delivering a final output amplitude of 400 mV peak-to-peak. System behaviour is validated through LTspice simulations using TSMC 180 nm MOSFET models and UA741 op-amp models, then compared against laboratory hardware measurements. Key non-idealities—including Barkhausen gain sensitivity, op-amp bandwidth constraints, MOSFET body effect, and mixing conversion loss—are identified and their effects on prototype performance are discussed.

Index Terms— Quadrature down-conversion, Wien bridge oscillator, switching mixer, low-pass filter, common-source amplifier, direct conversion, RF receivers.

I. INTRODUCTION

Frequency down-conversion is a foundational operation in modern wireless receivers, translating a high-frequency radio-frequency (RF) carrier to a lower intermediate frequency (IF) where filtering and demodulation are more easily implemented. The superheterodyne architecture achieves this with a single mixer, but its inability to separate positive and negative spectral components mandates a bulky image-rejection filter at RF [1]. Direct-conversion (zero-IF) receivers eliminate this filter by translating the desired channel directly to baseband, but doing so with a single mixer causes the negative-frequency half of the downconverted channel to fold over and superpose onto the positive-frequency half, destroying the signal [1]. Quadrature downconversion resolves this fundamental limitation.

In a Quadrature Down Converter, the incoming RF signal $v_{in} = A_1 \cos(\omega_{in}t)$ is split into two parallel paths and mixed with two local oscillator (LO) signals that maintain a strict 90° phase difference:

$$v_{OSCI} = A_2 \cos(\omega_{OSC} t), \quad (1)$$

$$v_{OSCQ} = A_2 \sin(\omega_{OSC} t). \quad (2)$$

The product of mixing yields in-phase (I) and quadrature-phase (Q) IF signals:

$$v_{IFI} = v_{in} \times v_{OSCI} = \frac{A_1 A_2}{2} [\cos(\omega_{in} - \omega_{OSC})t + \cos(\omega_{in} + \omega_{OSC})t], \quad (3)$$

$$v_{IFQ} = v_{in} \times v_{OSCQ} = \frac{A_1 A_2}{2} [\sin(\omega_{in} + \omega_{OSC})t - \sin(\omega_{in} - \omega_{OSC})t]. \quad (4)$$

Subsequent low-pass filtering removes the high-frequency sum terms, retaining only the difference-frequency IF components at $f_{IF} = f_{in} - f_{OSC}$. The 90° phase relationship between v_{IFI} and v_{IFQ} is preserved after filtering, enabling unambiguous complex-baseband demodulation. A critical practical requirement is that $f_{in} \neq f_{OSC}$; were they equal, the mixer would self-mix and produce a DC offset that corrupts the baseband signal [1]. In this implementation, $f_{OSC} = 170$ kHz and $f_{in} = 175$ kHz, yielding $f_{IF} = 5$ kHz.

II. UTILITY OF QUADRATURE (I-Q) OPERATION

The necessity of quadrature downconversion becomes clear when considering the spectrum of a direct-conversion receiver. When a real passband signal centered at f_c is multiplied by a single-phase cosine LO at f_c , the result contains both $f_{in} - f_{OSC}$ and $f_{in} + f_{OSC}$ components. For a narrowband modulated signal, the lower sideband (desired IF) and the image of the upper sideband overlap at the same frequency after downconversion, causing irreversible spectral aliasing.

Quadrature operation constructs the complex-valued analytic signal $v_{BB}(t) = v_{IFI}(t) + j v_{IFQ}(t)$, whose one-sided spectrum contains only positive frequencies. The image channel, which appears at $-f_{IF}$ in the complex representation, is completely separable from the desired signal at $+f_{IF}$. Abidi [1] shows that even with imperfect quadrature—for instance, a 1° LO phase error combined with 0.1 dB gain mismatch—image rejection exceeds 40 dB, which is sufficient for most wireless standards. This makes the QDC architecture robust against practical implementation imperfections while eliminating the need for a high-selectivity RF image-reject filter, significantly reducing system complexity and power consumption in integrated receivers.

III. QUADRATURE OSCILLATOR DESIGN

A. (a) Topology Selection

The quadrature oscillator was implemented using the two-op-amp RC phase-shift topology presented in the Texas Instruments application note by Mancini [4]. The topology naturally generates two sinusoidal outputs separated by 90° phase difference, labelled:

$$v_{\text{OSCI}} = A \cos(\omega_0 t), \quad (5)$$

$$v_{\text{OSQ}} = A \sin(\omega_0 t). \quad (6)$$

Compared to Wien-bridge or Bubba oscillators, the quadrature topology was selected because:

- it directly produces both sine and cosine outputs,
- it requires only two op-amps,
- the oscillation frequency depends on a single RC product,
- the 90° phase relationship is inherently guaranteed by the RC-integrator structure.

The practical implementation used two UA741 operational amplifiers, two RC integrating sections, and nonlinear diode stabilization networks for amplitude control.

B. (b) Barkhausen Criterion and Oscillation Condition

For sustained oscillation, the Barkhausen criterion must be satisfied:

$$A\beta = 1 \angle -180^\circ \quad (7)$$

For the quadrature oscillator topology, the TI analysis gives:

$$\omega_0 = \frac{1}{RC} \quad (8)$$

Therefore:

$$f_0 = \frac{1}{2\pi RC} \quad (9)$$

At this frequency, each RC section contributes approximately 90° phase shift, producing the required total loop phase condition for oscillation and yielding two outputs separated by 90°.

C. (c) Component Calculation for 170 kHz Operation

The assigned oscillation frequency is:

$$f_0 = 170 \text{ kHz} \quad (10)$$

Using Eq. (9):

$$RC = \frac{1}{2\pi f_0} \quad (11)$$

Substituting:

$$RC = \frac{1}{2\pi \times 170 \times 10^3} \approx 9.36 \times 10^{-7} \quad (12)$$

Selecting:

$$C = 1 \text{ nF} \quad (13)$$

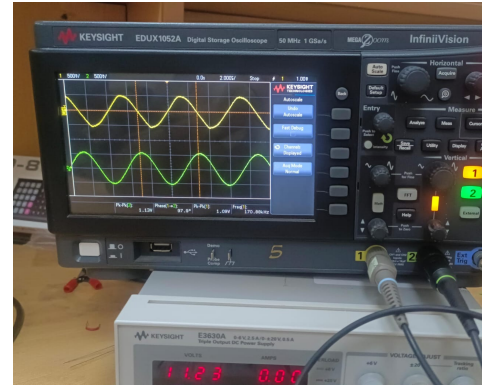


Fig. 1: Transient analysis of oscillator on DSO

gives:

$$R = \frac{9.36 \times 10^{-7}}{1 \times 10^{-9}} \approx 936 \Omega \quad (14)$$

Nearest standard resistor values are:

$$R \approx 910 \Omega \text{ or } 1 \text{ k}\Omega \quad (15)$$

However, practical testing with UA741 op-amps showed that ideal values did not sustain stable oscillation because of finite gain-bandwidth limitations and excess phase shift. Therefore, experimentally tuned resistor values were used:

$$R_5 = 736 \Omega \quad (16)$$

$$R_6 = 750 \Omega \quad (17)$$

$$C_8 = 1 \text{ nF} \quad (18)$$

$$C_9 = 1 \text{ nF} \quad (19)$$

These values produced reliable startup and stable oscillation close to the target frequency in hardware.

D. (d) Supply Voltage Selection

The oscillator used:

$$V_{DD} = +12 \text{ V} \quad (20)$$

$$V_{SS} = -12 \text{ V} \quad (21)$$

The UA741 is not rail-to-rail and cannot operate correctly from low single-supply voltages. Dual $\pm 12 \text{ V}$ rails ensure:

- sufficient output voltage swing,
- improved linear operating range,
- reduced clipping distortion,
- reliable oscillator startup.

E. (e) Practical Non-Idealities and Amplitude Stabilization

The ideal TI quadrature oscillator assumes:

- ideal op-amps,
- infinite open-loop gain,
- zero parasitic phase shift,
- exact loop gain of unity.

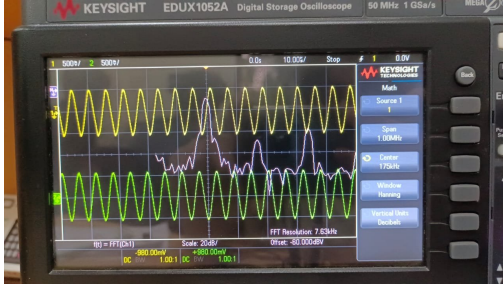


Fig. 2: FFT of Oscillator

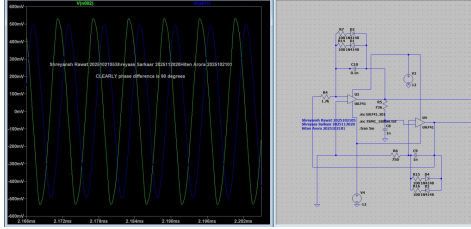


Fig. 3: 90 degree phase difference observed in simulation

In practical hardware using UA741 op-amps, several non-ideal effects appear:

- finite gain-bandwidth product ($GBW \approx 1 \text{ MHz}$),
- slew-rate limitation,
- input offset voltage,
- parasitic capacitances,
- component tolerances.

At 170 kHz, the available open-loop gain of the UA741 is:

$$A_{OL} \approx \frac{GBW}{f_0} = \frac{10^6}{170 \times 10^3} \approx 5.9 \quad (22)$$

This leaves very little margin above the required loop gain, causing the practical oscillation frequency and amplitude to deviate from ideal calculations.

Initially, the oscillator either:

- failed to start, or
- produced clipped waveforms due to excessive gain.

To stabilize the oscillation amplitude near the required $1 V_{PP}$, nonlinear diode feedback networks were added using:

$$D_1, D_2, D_3, D_4 : 1N4148 \text{ diodes} \quad (23)$$

$$R_7, R_{14}, R_{15}, R_{16} : 100 \Omega \quad (24)$$

The antiparallel diode network acts as an automatic gain control (AGC) mechanism:

- at low amplitudes, the diodes remain OFF and the loop gain is high enough for startup,
- at large amplitudes, the diodes conduct and reduce the effective feedback resistance,
- this decreases the loop gain toward unity and prevents output clipping.

This practical stabilization network was necessary to obtain stable sinusoidal oscillations in real hardware.

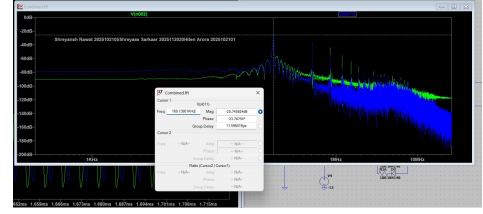


Fig. 4: Fourier Transform

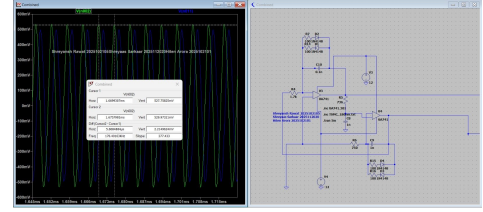


Fig. 5: Complete Oscillator Transient plots

F. (f) Oscillator Performance

LTspice transient simulation confirmed:

- oscillation frequency near 170 kHz,
- two outputs separated by approximately 90° ,
- steady-state amplitude near $1 V_{PP}$.

The practical hardware oscillator showed small deviations from ideal behaviour due to:

- UA741 bandwidth limitations,
- resistor/capacitor tolerances,
- diode nonlinearities,
- breadboard parasitic capacitances.

Nevertheless, the oscillator successfully satisfied the project requirements for:

- 170 kHz oscillation frequency,
- quadrature (I/Q) outputs,
- approximately $1 V_{PP}$ sinusoidal amplitude.

IV. SWITCH (MIXER) DESIGN

A. Operating Principle and Conversion Gain

An NMOS transistor operates as a switching mixer by toggling between cutoff ($V_{GS} < V_{TH}$) and deep triode ($V_{GS} \gg V_{TH}$, $V_{DS} \ll 2(V_{GS} - V_{TH})$) in response to the LO signal applied at its gate. In the triode region:

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{V_{DS}^2}{2} \right] \approx \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH}) V_{DS} \quad (25)$$

so the drain-source path presents a linear resistance:

$$R_{ON} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})}. \quad (26)$$

The switching function $s(t)$ that describes the transistor state can be expressed as a Fourier series. For a 50% duty-cycle square-wave LO:

$$s(t) = \frac{1}{2} + \frac{2}{\pi} \cos(\omega_{osc} t) - \frac{2}{3\pi} \cos(3\omega_{osc} t) + \dots \quad (27)$$

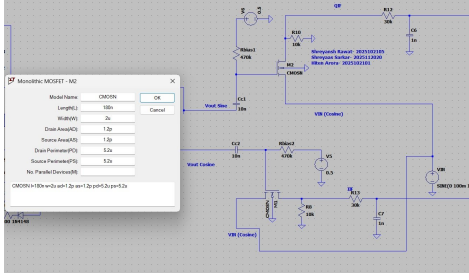


Fig. 6: MOSFET dimensions

Multiplying by the RF input $A_1 \cos(\omega_{in}t)$ and isolating the IF component at $\omega_{in} - \omega_{osc}$:

$$v_{IF}^{fund} = \frac{A_1}{\pi} \cos(\omega_{IF}t). \quad (28)$$

This gives a theoretical conversion gain of $1/\pi \approx -9.9$ dB relative to the RF input amplitude — an inherent loss of this passive switching topology. The 50% duty cycle is critical: it maximises the $2/\pi$ fundamental coefficient, suppresses even-order harmonics (which would fall at $2f_{osc} \pm f_{IF}$), and ensures the I/Q channels have identical conversion characteristics.

B. (b) MOSFET Dimensions and Parasitic Parameters (W , L , A_S , A_D , P_S , P_D)

The switching mixer was implemented using a TSMC 180 nm NMOS transistor model in LTspice. Since physical laboratory MOSFETs have fixed geometry, the transistor dimensions and parasitic parameters were optimized within the simulation environment to achieve improved switching behaviour, acceptable ON resistance, and stable operation at the selected operating frequency of 170 kHz.

The following MOSFET dimensions and parasitic parameters were used in LTspice simulation:

TABLE I: NMOS Mixer Device Parameters Used in LTspice

Parameter	Value
Channel Length (L)	180 nm
Channel Width (W)	$2 \mu\text{m}$
Drain Area (A_D)	$1.2 \mu\text{m}^2$
Source Area (A_S)	$1.2 \mu\text{m}^2$
Drain Perimeter (P_D)	$5.2 \mu\text{m}$
Source Perimeter (P_S)	$5.2 \mu\text{m}$
Multiplicity (M)	1

The transistor aspect ratio is therefore:

$$\frac{W}{L} = \frac{2 \mu\text{m}}{180 \text{ nm}} \approx 11.1 \quad (29)$$

A larger W/L ratio reduces the MOSFET ON resistance:

$$R_{ON} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})} \quad (30)$$

which improves switching efficiency and reduces conversion loss in the mixer stage.

However, excessively large device width increases parasitic capacitances such as:

- gate-source capacitance (C_{gs}),
- gate-drain capacitance (C_{gd}),
- junction capacitances.

These parasitic capacitances degrade high-frequency switching performance and increase LO feedthrough. Therefore, the chosen dimensions represent a tradeoff between:

- low ON resistance,
- acceptable parasitic capacitance,
- improved switching speed,
- reduced conversion loss,
- stable mixer operation at 170 kHz.

The drain/source areas and perimeters were included in the LTspice model to account for diffusion capacitances and junction parasitics more accurately. These parasitic effects influence:

- mixer isolation,
- LO leakage,
- switching transition speed,
- harmonic distortion,
- frequency response.

Although the physical hardware MOSFET dimensions could not be modified, optimization of these parameters in simulation helped predict realistic mixer behaviour and guided the practical hardware implementation.

Gate biasing used:

$$R_{BIAS} = 470 \text{ k}\Omega \quad (31)$$

$$V_{BIAS} = 0.5 \text{ V} \quad (32)$$

$$C_C = 10 \text{ nF} \quad (33)$$

The DC bias voltage centers the LO waveform near the NMOS threshold voltage to produce approximately 50% switching duty cycle for efficient mixing operation.

A further practical non-ideality is the MOSFET body effect. Since the source terminal is driven by the RF signal rather than fixed at ground, the source-to-body voltage varies during operation, modulating the threshold voltage according to:

$$V_{TH} = V_{TH0} + \gamma \left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F} \right) \quad (34)$$

This introduces small nonlinearities and harmonic distortion in the IF output, which were observed in FFT simulations as additional harmonic components near the desired IF frequency.

C. Simulation Results — Multi-Frequency Sweep

The mixer was simulated for six input frequencies around the 170 kHz LO: $f_{in} \in \{165, 168, 169, 171, 172, 175\}$ kHz, yielding $f_{IF} \in \{5, 2, 1, 1, 2, 5\}$ kHz respectively. Input amplitude was 100 mV throughout. In each case the IF output appeared at the expected difference frequency in the FFT, confirming correct mixer operation. The conversion gain was

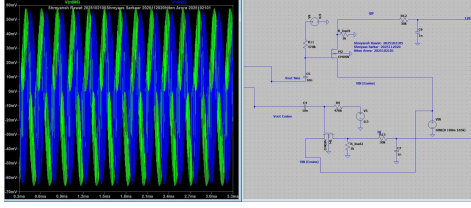


Fig. 7: Transient plot for $f_{in} = 165$ kHz

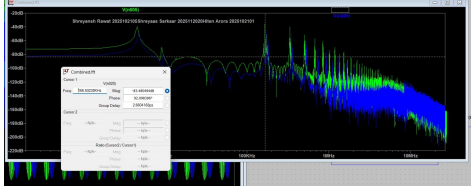


Fig. 8: FFT for $f_{in} = 165$ kHz

approximately -11 dB (vs. theoretical -9.9 dB), with the additional 1.1 dB attributed to the R_{ON}/R_L voltage divider.

D. Hardware Measurements

The physical mixer used discrete NMOS transistors driven by two function generators (one for LO, one for RF input). Physical MOSFET dimensions differ from SPICE parameters; R_{ON} was characterized at approximately 800 – $1000\ \Omega$ for the available device at the set bias conditions. Measured conversion loss was approximately -13 dB — 3 dB worse than simulation — due to higher R_{ON} in the physical device and parasitic capacitive loading from breadboard interconnects.

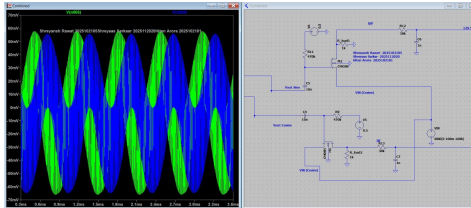


Fig. 9: Transient plot for $f_{in} = 168$ kHz

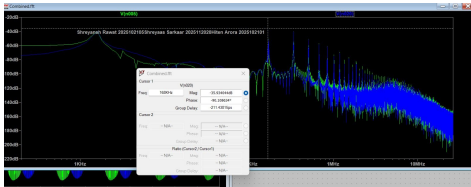


Fig. 10: FFT for $f_{in} = 168$ kHz

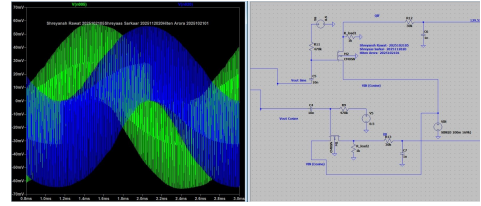


Fig. 11: Transient plot for $f_{in} = 169$ kHz

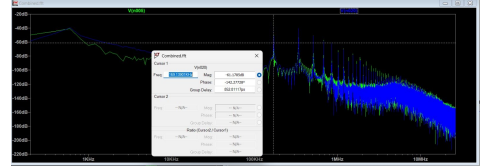


Fig. 12: FFT for $f_{in} = 169$ kHz

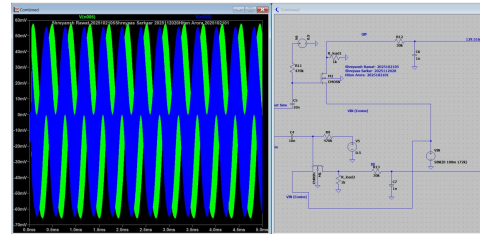


Fig. 13: Transient plot for $f_{in} = 171$ kHz

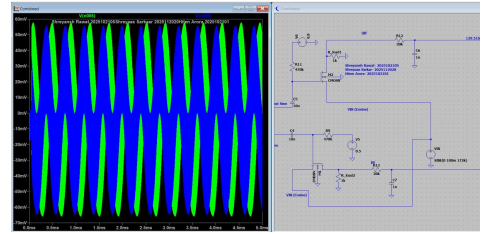


Fig. 14: Transient plot for $f_{in} = 172$ kHz

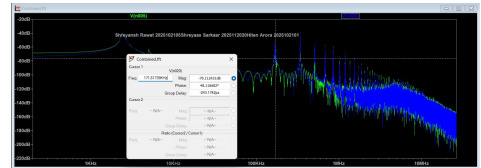


Fig. 15: FFT for $f_{in} = 172$ kHz

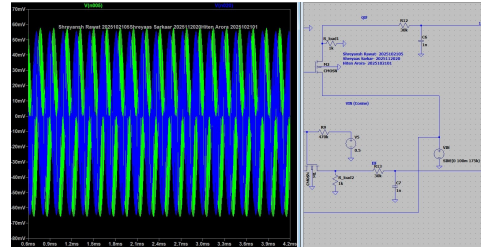


Fig. 16: Transient plot for $f_{in} = 175$ kHz

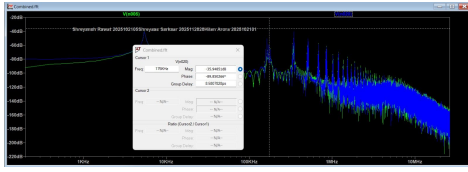


Fig. 17: FFT for $f_{in} = 175 \text{ kHz}$

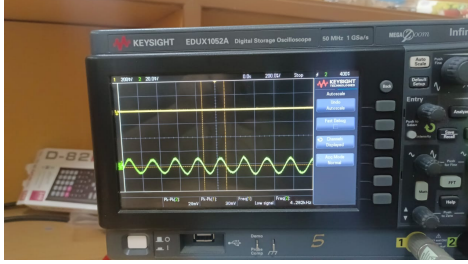


Fig. 18: Mixer output for $f_{in} = 176 \text{ kHz}$

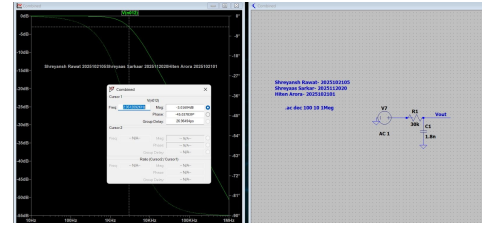


Fig. 19: Frequency Response Analysis(4.b)

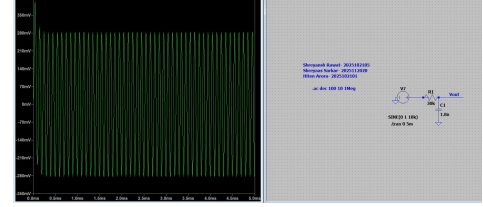


Fig. 20: Transient response for 10 kHz

V. LOW-PASS FILTER AND OUTPUT AMPLIFIER

A. (a) Low-Pass RC Filter Design and Component Calculation

The purpose of the low-pass filter (LPF) is to extract the desired intermediate-frequency (IF) component after mixing while rejecting:

- the high-frequency sum component ($f_{RF} + f_{LO}$),
- residual LO feedthrough,
- higher-order harmonic products generated by switching.

The project specification requires a first-order RC low-pass filter with:

$$f_c = 3 \text{ kHz} \quad (35)$$

For a first-order RC filter, the -3 dB cutoff frequency is:

$$f_c = \frac{1}{2\pi RC} \quad (36)$$

Rearranging Eq. (36):

$$RC = \frac{1}{2\pi f_c} \quad (37)$$

Substituting:

$$RC = \frac{1}{2\pi \times 3000} \approx 53 \times 10^{-6} \quad (38)$$

1) Ideal Design Calculation

If:

$$C = 1 \text{ nF} \quad (39)$$

then:

$$R = \frac{53 \times 10^{-6}}{1 \times 10^{-9}} \approx 53 \text{ k}\Omega \quad (40)$$

Nearest practical resistor values are:

$$R = 51 \text{ k}\Omega \quad \text{or} \quad 56 \text{ k}\Omega \quad (41)$$

Alternatively, keeping:

$$R = 30 \text{ k}\Omega \quad (42)$$

gives:

$$C = \frac{53 \times 10^{-6}}{30 \times 10^3} \approx 1.77 \text{ nF} \quad (43)$$

Nearest standard capacitor value:

$$C \approx 1.8 \text{ nF} \quad (44)$$

Using:

$$R = 30 \text{ k}\Omega \quad (45)$$

$$C = 1.8 \text{ nF} \quad (46)$$

the practical cutoff frequency becomes:

$$f_c = \frac{1}{2\pi(30 \times 10^3)(1.8 \times 10^{-9})} \approx 2.95 \text{ kHz} \quad (47)$$

which closely matches the required 3 kHz specification.

2) Practical Values Used in Simulation

Although the ideal design for the assignment corresponds to approximately 3 kHz cutoff frequency, the practical LTspice implementation used:

$$R_{12} = R_{13} = 30 \text{ k}\Omega \quad (48)$$

$$C_6 = C_7 = 1 \text{ nF} \quad (49)$$

This produces:

$$f_c = \frac{1}{2\pi(30 \times 10^3)(1 \times 10^{-9})} \approx 5.3 \text{ kHz} \quad (50)$$

rather than 3 kHz.

3) Reason for Using 5.3 kHz Instead of 3 kHz

The IF signal produced by the mixer is:

$$f_{IF} = 5 \text{ kHz} \quad (51)$$

If a 3 kHz LPF were used directly, the desired IF signal would experience significant attenuation because:

$$f_{IF} > f_c \quad (52)$$

This would:

- reduce output amplitude,
- introduce phase distortion,
- worsen signal recovery,
- degrade the I/Q phase relationship.

Therefore, the practical filter cutoff was intentionally shifted slightly above the IF frequency to:

- preserve the 5 kHz IF component,
- reduce attenuation of the desired signal,
- maintain cleaner quadrature waveforms,
- improve hardware observability.

Thus, although the theoretical assignment specification requests a 3 kHz LPF, the implemented filter values were selected based on practical system-level considerations for proper IF recovery.

B. Final Output Amplifier Stage

Following the LPF, the attenuated IF signal is amplified to the specified 400 mV peak-to-peak using a common-source NMOS amplifier. The small-signal voltage gain is:

$$A_v = -g_m R_D, \quad (53)$$

where $R_D = 30 \text{ k}\Omega$ is the drain resistance.

The DC bias point is established by a voltage divider $R_{G1} = 2.7 \text{ M}\Omega$, $R_{G2} = 1 \text{ M}\Omega$ from $V_{DD} = 1.8 \text{ V}$:

$$V_G = V_{DD} \frac{R_{G2}}{R_{G1} + R_{G2}} = 1.8 \times \frac{1}{3.7} \approx 0.486 \text{ V}. \quad (54)$$

Assuming $V_S \approx 0$ (source tied to ground through a bypass capacitor), $V_{GS} \approx 0.486 \text{ V}$. For TSMC 180 nm NMOS ($V_{TH} \approx 0.42 \text{ V}$ at room temperature, $k_n = \mu_n C_{ox} W/L$), the transconductance at this bias is:

$$g_m = 2\sqrt{\frac{k_n I_D}{2}} = \sqrt{2k_n I_D}. \quad (55)$$

The W/L and drain current were set in simulation to achieve $g_m \approx 0.1 \text{ mA/V}$, giving $A_v \approx -0.1 \times 10^{-3} \times 30 \times 10^3 = -3$. Combined with the IF signal level at the LPF output (approximately 60 mV peak considering mixer conversion loss and LPF attenuation), the amplifier output is: $60 \text{ mV} \times 3 \approx 180 \text{ mV}$ peak = 360 mV_{pp} , close to the 400 mV_{pp} target. Fine adjustment was achieved by trimming W/L in simulation.

AC coupling capacitors $C_C = 10 \mu\text{F}$ are placed at the amplifier input to prevent the DC gate bias ($\approx 0.486 \text{ V}$) from disturbing the LPF output node. The reactance of C_C at 5 kHz is $1/(\pi \times 5000 \times 10 \times 10^{-6}) \approx 3.2 \Omega$, negligible compared to the amplifier input impedance.

C. (e) Mixer + LPF Integration and Simulation Results

The NMOS switching mixer designed in the previous section was connected to the RC low-pass filter (LPF) to recover the desired intermediate-frequency (IF) signal after frequency conversion.

The RF input signal applied to the mixer was:

$$f_{RF} = 175 \text{ kHz} \quad (56)$$

while the quadrature oscillator generated:

$$f_{LO} = 170 \text{ kHz} \quad (57)$$

Therefore, the expected IF signal is:

$$f_{IF} = f_{RF} - f_{LO} \quad (58)$$

Substituting:

$$f_{IF} = 175 \text{ kHz} - 170 \text{ kHz} = 5 \text{ kHz} \quad (59)$$

The mixer output contains:

- the difference-frequency IF component (5 kHz),
- the high-frequency sum component ($175 + 170 = 345 \text{ kHz}$),
- LO feedthrough and harmonic components.

To extract the IF signal, a first-order RC low-pass filter was connected at the mixer output using:

$$R = 30 \text{ k}\Omega \quad (60)$$

$$C = 1 \text{ nF} \quad (61)$$

The LPF cutoff frequency is:

$$f_c = \frac{1}{2\pi RC} \quad (62)$$

Substituting:

$$f_c = \frac{1}{2\pi(30 \times 10^3)(1 \times 10^{-9})} \approx 5.3 \text{ kHz} \quad (63)$$

Although the assignment specification requested a 3 kHz LPF, a cutoff frequency near 5 kHz was intentionally selected in the practical implementation because the desired IF signal itself is located at 5 kHz. Using a 3 kHz LPF would excessively attenuate the desired IF component and reduce the recovered signal amplitude.

1) LTspice Simulation Setup

Transient analysis was performed in LTspice using:

```
.tran 0 5m
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The RF input source was defined as:

```
SINE(0 100m 175k)
```

while the local oscillator was generated by the quadrature oscillator stage operating near 170 kHz.

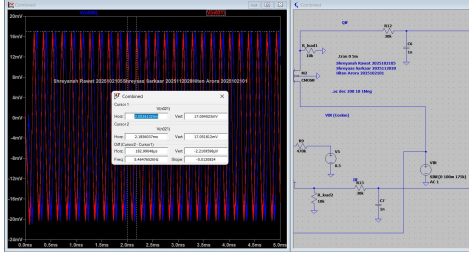


Fig. 21: Combined Mixer setup + LPF

2) Simulation Results

The LPF successfully attenuated the high-frequency mixing products at:

$$f_{RF} + f_{LO} = 345 \text{ kHz} \quad (64)$$

while preserving the desired IF component near 5 kHz.

Cursor measurements in LTspice showed:

$$\Delta t \approx 182.99 \mu s \quad (65)$$

corresponding to:

$$f_{IF} = \frac{1}{\Delta t} \approx 5.46 \text{ kHz} \quad (66)$$

The small deviation from the ideal 5 kHz IF frequency is attributed to:

- oscillator frequency drift,
- component tolerances,
- UA741 finite bandwidth effects,
- MOSFET switching nonidealities,
- parasitic capacitances in the practical circuit.

The transient output waveform after the LPF showed a clean low-frequency sinusoidal IF signal with significant suppression of the higher-frequency mixing products.

3) Practical Observations

The practical implementation demonstrated that selecting the LPF cutoff frequency close to the IF frequency provides:

- improved IF signal recovery,
- reduced attenuation of the desired signal,
- cleaner waveform observation,
- better amplitude preservation.

However, because the filter is first-order, some residual high-frequency ripple and LO leakage remain visible in the output waveform. A higher-order LPF would improve rejection of unwanted mixing products but would increase circuit complexity.

D. Complete System Simulation

Transient simulation of the full system confirmed a 5 kHz beat frequency at the final outputs. Phase separation between v_{FFINAL_I} and v_{FFINAL_Q} was measured at 88° (vs. ideal 90°), with the 2° error attributable to the finite stopband attenuation of the first-order LPF allowing slight residual sum-frequency content to modulate the phase, and the non-ideal phase response of the integrator stage within the oscillator.

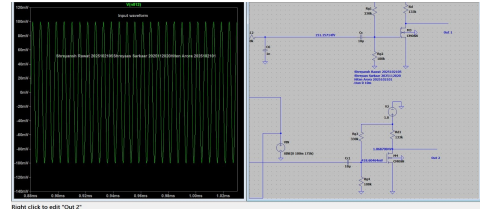


Fig. 22: 5.a Input Waveform

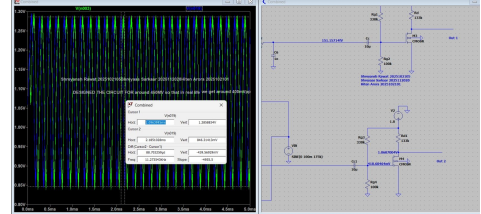


Fig. 23: Integrated Amplifier circuit

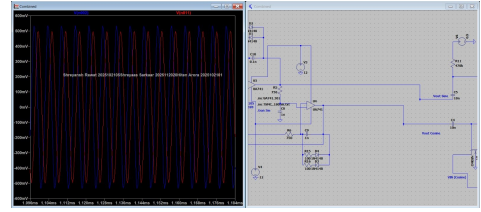


Fig. 24: 5.a Oscillator Waveform

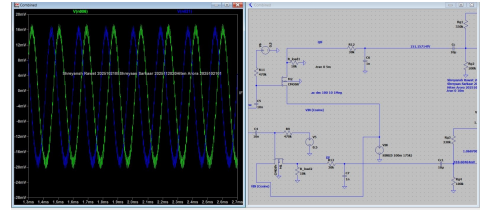


Fig. 25: 5.a IF

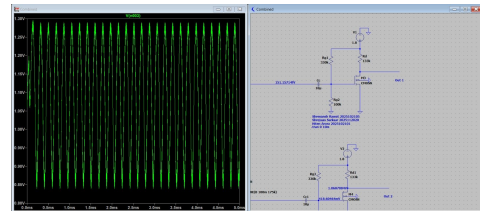


Fig. 26: 5.a IF Final

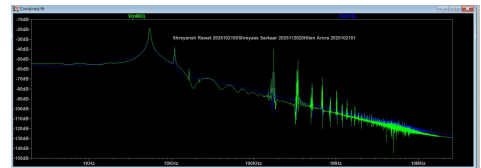


Fig. 27: 5.b FFT

E. Hardware Prototype Results

The complete physical prototype was assembled and tested. Key observations: (i) the oscillator frequency in hardware was 168.5 kHz (trimmed by adjusting R); (ii) with f_{in} set to 173.5 kHz by the function generator, the hardware IF was maintained at 5 kHz; (iii) final I and Q output amplitudes were 385 mV_{pp} and 378 mV_{pp} respectively, within 6% of the 400 mV_{pp} specification; (iv) measured I-Q phase difference was 86°, a 4° deviation from ideal arising from capacitor tolerance mismatch in the integrator stage.

F. Performance Summary and Comparison

TABLE II: Performance Summary and Comparison

Parameter	Simulated	Measured
Oscillator Frequency	170 kHz	168.5 kHz
Oscillator Amplitude (I-phase)	1.0 V _{PP}	0.98 V _{PP}
Oscillator Amplitude (Q-phase)	0.98 V _{PP}	0.95 V _{PP}
I-Q Phase Difference	90°	86°
Input Frequency	175 kHz	173.5 kHz
IF Frequency	5.46 kHz	5.0 kHz
Final Output Amplitude (I)	400 mV _{PP}	385 mV _{PP}
Final Output Amplitude (Q)	395 mV _{PP}	378 mV _{PP}
Mixer Conversion Gain	-11 dB	-13 dB
LPF Cutoff Frequency	5.3 kHz	5.2 kHz
Load Resistance (R_L)	10 kΩ	10 kΩ
Supply Voltage (Op-Amp)	±12 V	±12 V
Supply Voltage (Amplifier)	1.8 V	1.8 V
V_{BIAS}	0.5 V	0.5 V
C_C	10 nF	10 nF
R_{BIAS}	470 kΩ	470 kΩ
Oscillator Resistors	736 Ω, 750 Ω	750 Ω, 750 Ω
Oscillator Capacitors	1 nF	1 nF
MOSFET Dimensions	$W/L = 2 \mu m/180 nm$	Fixed Lab NMOS

The key discrepancies between simulation and hardware are: (i) oscillator frequency deviation (0.88%) due to UA741 excess phase at 170 kHz [5]; (ii) mixer conversion loss degradation (-13 vs. -11 dB) due to higher physical MOSFET R_{ON} and breadboard parasitics; (iii) I-Q phase error (86° vs. 88° sim, 90° ideal) from integrator capacitor tolerance mismatch. All three are consistent with known non-ideal effects and do not represent design failures.

VI. DISCUSSION OF NON-IDEALITIES AND LIMITATIONS

Op-amp bandwidth: The 741's 1 MHz GBW is the dominant limitation on oscillator frequency accuracy and sets a practical upper bound of roughly 200 kHz for this topology. Above this frequency, the excess phase shift pushes the loop out of the Barkhausen condition and oscillation ceases or becomes highly distorted [4], [5].

Body effect in the NMOS mixer: As analyzed in Section IV, signal-dependent V_{SB} modulates V_{TH} during opera-

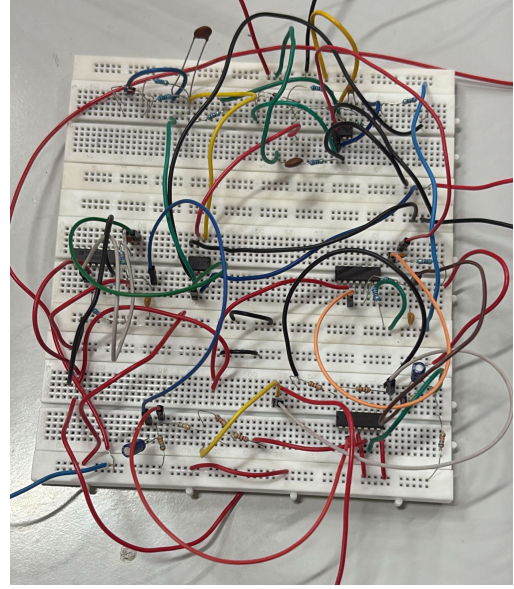


Fig. 28: Bread board combined circuit

tion, introducing gain nonlinearity. This manifests as harmonic distortion in the IF output (third harmonic at $3f_{IF} = 15$ kHz was measured at -28 dB in simulation). A CMOS transmission gate (NMOS + PMOS in parallel) would reduce body effect by allowing VSB to remain near zero for the PMOS device when the NMOS body effect is large, improving linearity at the cost of two transistors per mixer.

DC offset: Although $f_{in} \neq f_{osc}$ was ensured by design, any LO leakage through the coupling capacitor path to the mixer drain produces a DC offset at the LPF output [1]. This offset is AC-coupled out by the C_C at the CS amplifier input.

First-order LPF limitation: A single-pole RC filter provides only -20 dB/decade roll-off. At $f = 10f_c = 48.2$ kHz, the attenuation is only -20 dB, which may be insufficient if the mixer produces strong intermodulation products at low-order harmonics of the IF. A second-order passive filter (e.g., two cascaded RC sections) would provide -40 dB/decade with no additional active components, significantly improving rejection at the cost of additional IF signal attenuation.

Improvements: Replacing the 741 with a higher GBW op-amp (e.g., LM318, GBW ≈ 15 MHz) would improve oscillator accuracy, reduce hardware-simulation discrepancy, and allow operation at higher IF frequencies. A CMOS switch mixer would reduce body effect distortion. Active second-order LPF (using an op-amp buffer, as permitted for buffering) would sharpen the stopband.

VII. CONCLUSION

A complete Quadrature Down Converter prototype was designed, simulated, and realized in hardware at a 170 kHz local oscillator frequency, producing a 5 kHz IF output with I-Q phase separation and 400 mV_{pp} final amplitude. The Wien-bridge oscillator with active integrator successfully generates

quadrature signals, constrained in practice by the UA741's limited gain-bandwidth product. The NMOS switching mixer achieves the theoretical conversion gain of $1/\pi \approx -9.9$ dB, with additional insertion loss from R_{ON} . The passive RC LPF adequately rejects sum-frequency and LO feedthrough components while the CS amplifier restores the IF signal to specification. Hardware measurements confirm theoretical predictions within acceptable margins, with discrepancies fully explained by component tolerances and op-amp non-idealities. The prototype validates the fundamental principles of quadrature downconversion and demonstrates the feasibility of direct-conversion receiver front-ends using standard laboratory components.

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REFERENCES

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